

Notes on a staged QUICK–MBX interface for self-consistent QM/MB-pol

QUICK development notes

June 21, 2026

1 Scientific target

The first target is a CPU-only, energy-only, fully self-consistent QM/MB-pol coupling between QUICK and MBX. QUICK owns the quantum density, AO operators, and ESP/EFIELD/EFG evaluation. MBX owns MB-pol/MB-nrg many-body terms, permanent electrostatics, polarizabilities, and induced dipoles. The immediate implementation order is serial CPU first and MPI CPU second. GPU, gradients, and MD are later stages.

The main references for the design are Lambros et al. (2020), “A many-body, fully polarizable approach to QM/MM simulations”, OpenMMPol (2024), and the MBX source/documentation. The OpenMMPol paper is the best architectural reference for a clean separation between a QM engine and a polarizable MM engine, while Lambros et al. define the MB-pol QM/MM scientific target.

2 QM/MB-MM theory

For fixed nuclear coordinates, partition the system into a QM subsystem Q and a many-body molecular-mechanics subsystem M . The additive polarizable QM/MB-MM energy can be written as

$$\mathcal{E}(\mathbf{P}, \boldsymbol{\mu}) = E_{\text{QM}}(\mathbf{P}) + E_{\text{MB-MM}}^{\text{internal}} + E_{\text{QM/MB-MM}}^{\text{perm}}(\mathbf{P}) + E_{\text{pol}}(\mathbf{P}, \boldsymbol{\mu}) + E_{\text{QM/MB-MM}}^{\text{sr/disp}}. \quad (1)$$

Here $\mathbf{P}$ is the QM density matrix and $\boldsymbol{\mu}$ is the vector of MB-MM induced dipoles. The term $E_{\text{MB-MM}}^{\text{internal}}$ contains the purely MB-MM energy terms within M ; the last three terms are the QM/MB-MM coupling terms. For MB-pol/MB-nrg, the internal term is itself a many-body potential containing one-body distortions, permanent electrostatics, polarization, dispersion, and short-range polynomial corrections between MB-MM monomers.

2.1 QM electrostatic potential and field

For an MB-MM electrostatic site k at position $\mathbf{R}_k$, with QM nuclei A at positions $\mathbf{R}_A$, the QM electrostatic potential is

$$\Phi_k^{\text{QM}}(\mathbf{P}) = \sum_{A \in Q} \frac{Z_A}{|\mathbf{R}_k - \mathbf{R}_A|} - \int \frac{\rho_{\mathbf{P}}(\mathbf{r})}{|\mathbf{R}_k - \mathbf{r}|} d\mathbf{r}. \quad (2)$$

The corresponding electric field in the QUICK convention is

$$\mathbf{E}_k^{\text{QM}}(\mathbf{P}) = \sum_{A \in Q} Z_A \frac{\mathbf{R}_k - \mathbf{R}_A}{|\mathbf{R}_k - \mathbf{R}_A|^3} - \int \rho_{\mathbf{P}}(\mathbf{r}) \frac{\mathbf{R}_k - \mathbf{r}}{|\mathbf{R}_k - \mathbf{r}|^3} d\mathbf{r}. \quad (3)$$

Thus $\mathbf{E}^{\text{QM}} = -\nabla_{\mathbf{R}}\Phi^{\text{QM}}$, and the field-gradient tensor printed by QUICK is $G_{ij} = \partial E_i / \partial R_j$.

2.2 Permanent electrostatic coupling

The permanent electrostatic QM/MB-MM cross energy is

$$E_{\text{QM/MB-MM}}^{\text{perm}}(\mathbf{P}) = \sum_{k \in M} q_k \Phi_k^{\text{QM}}(\mathbf{P}) = \mathbf{q}^\dagger \Phi^{\text{QM}}(\mathbf{P}). \quad (4)$$

This expression contains both QM nuclear–MB charge and QM electronic-density–MB charge interactions. In an AO basis, the electronic part contributes the usual point-charge embedding operator

$$h_{\mu\nu}^{\text{perm}} = - \sum_{k \in M} q_k \int \frac{\chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r})}{|\mathbf{r} - \mathbf{R}_k|} d\mathbf{r}. \quad (5)$$

2.3 Polarization coupling

Let $\mathbf{E}^{\text{MB-MM}}$ be the electric field at the polarizable sites due to the permanent MB-MM charge distribution and let $\mathbf{T}$ be the Thole-damped induced-dipole interaction matrix, including the inverse polarizabilities on its diagonal blocks. The polarization functional is

$$E_{\text{pol}}(\mathbf{P}, \boldsymbol{\mu}) = \frac{1}{2} \boldsymbol{\mu}^\dagger \mathbf{T} \boldsymbol{\mu} - \boldsymbol{\mu}^\dagger [\mathbf{E}^{\text{MB-MM}} + \mathbf{E}^{\text{QM}}(\mathbf{P})]. \quad (6)$$

The induced dipoles are the minimizers of this functional and satisfy

$$\mathbf{T} \boldsymbol{\mu} = \mathbf{E}^{\text{MB-MM}} + \mathbf{E}^{\text{QM}}(\mathbf{P}). \quad (7)$$

At the stationary induced dipoles,

$$E_{\text{pol}}(\mathbf{P}, \boldsymbol{\mu}_*) = -\frac{1}{2} \boldsymbol{\mu}_*^\dagger [\mathbf{E}^{\text{MB-MM}} + \mathbf{E}^{\text{QM}}(\mathbf{P})]. \quad (8)$$

The factor of one half belongs to the stationary polarization energy, not to the QM Fock coupling. By the variational stationarity of $\boldsymbol{\mu}$, the electronic induced-dipole operator entering the QM SCF is the full derivative of the field term,

$$h_{\mu\nu}^{\text{ind}} = - \sum_{k \in M} \boldsymbol{\mu}_k \cdot \mathbf{E}_{\mu\nu}^{\text{elec}}(\mathbf{R}_k), \quad (9)$$

where $\mathbf{E}_{\mu\nu}^{\text{elec}}$ is the electronic electric-field operator generated by one AO density element. There is no extra one-half factor in this operator.

2.4 Many-body short-range and dispersion coupling

The many-body non-electrostatic correction should be a cross term between the QM and MB-MM regions, not the entire MB-MM internal energy. A compact way to write it is

$$E_{\text{QM/MB-MM}}^{\text{sr/disp}} = \sum_{\substack{S: S \cap Q \neq \emptyset \\ S \cap M \neq \emptyset}} V_S^{n\text{B}, \text{sr}} + \sum_{A \in Q} \sum_{k \in M} V_{Ak}^{\text{disp}}. \quad (10)$$

For an MB-pol water dimer, this is the cross two-body short-range polynomial plus the cross dispersion contribution. This term is the physical source of the short-range repulsive wall missing

from a purely electrostatic/polarization QM/MB-pol coupling. For the current validation driver, the MB-pol estimate of this residual is computed as

$$\Delta E_{\text{sr/disp}}^{\text{MB-pol}}(A, B) = E_{\text{MB-pol}}^{\text{full}}(A, B) - E_{\text{MB-pol}}^{\text{elec+pol}}(A, B), \quad (11)$$

where the second term is an MBX calculation on the same dimer geometry with the h2o/h2o two-body polynomial and dispersion disabled. The corrected validation energy is then

$$E_{A_{\text{QM}}/B_{\text{MB}}}^{\text{corr}} = E_{A_{\text{QM}}/B_{\text{MB}}}^{\text{elec+pol}} + \Delta E_{\text{sr/disp}}^{\text{MB-pol}}(A, B). \quad (12)$$

2.5 Water-dimer specialization

For the validation case with monomer A treated by QUICK and monomer B treated by MB-pol, the energy is

$$\begin{aligned} E_{A_{\text{QM}}/B_{\text{MB}}} &= E_{\text{QM}}^A(\mathbf{P}) + E_{\text{MB}}^{1\text{B}}(B) + \mathbf{q}_B^\dagger \Phi_A^{\text{QM}}(\mathbf{P}) \\ &\quad + \frac{1}{2} \boldsymbol{\mu}_B^\dagger \mathbf{T}_{BB} \boldsymbol{\mu}_B - \boldsymbol{\mu}_B^\dagger \left[\mathbf{E}_B^{\text{perm}} + \mathbf{E}_A^{\text{QM}}(\mathbf{P}) \right] \\ &\quad + V_{AB}^{2\text{B},\text{sr}} + V_{AB}^{\text{disp}}. \end{aligned} \quad (13)$$

Equations 4–9 define the energy-only self-consistent electrostatic/polarization target. The final two terms in Eq. 13 are the next correction needed to reproduce the short-range wall of the MB-pol dimer potential energy surface.

3 QUICK–MBX energy bookkeeping

At SCF iteration k , QUICK evaluates $\Phi^{\text{QM}}[\rho_k]$ and $\mathbf{E}^{\text{QM}}[\rho_k]$ at the MBX sites and sends them to MBX. MBX solves Eq. 7 and returns induced dipoles. QUICK then adds Eqs. 5 and 9 to the AO operator. Because the induced dipoles depend on the current density, QUICK’s delta-operator shortcut is disabled when MBX coupling is active.

The MBX adapter refreshes the MBX energy after the SCF density has converged by sending the final QUICK ESP/EFIELD to MBX and calling the MBX `Energy(false)` entry point. The final energy is not obtained by adding the raw MBX energy directly to QUICK, because the QUICK SCF energy already contains the electronic expectation value of the MBX permanent and induced one-electron operators.

Define the final QUICK SCF subtotal, before any MBX correction, as E_{Q}^{op} . This is the usual QUICK electronic plus nuclear energy evaluated with the MBX permanent-charge and induced-dipole operators present in the Fock matrix. Let

$$P = \sum_s q_s \Phi_s^{\text{QM}} = P_{\text{cl}} + P_{\text{el}}, \quad (14)$$

$$I = - \sum_s \boldsymbol{\mu}_s^{\text{ind}} \cdot \mathbf{E}_s^{\text{QM}} = I_{\text{cl}} + I_{\text{el}}, \quad (15)$$

where “cl” denotes the contribution from QUICK nuclei and any QUICK classical external charges, and “el” denotes the electronic-density contribution. Because E_{Q}^{op} already includes $P_{\text{el}} + I_{\text{el}}$, while MBX includes one half of the external permanent term and the variational polarization energy, the accepted stage-1.5 additive correction is

$$\Delta E_{\text{MBX}} = E_{\text{MBX}} - P_{\text{el}} - I_{\text{el}} + \frac{1}{2} P. \quad (16)$$

The printed total energy is therefore

$$E_{\text{tot}} = E_{\text{Q}}^{\text{op}} + \Delta E_{\text{MBX}}. \quad (17)$$

Equivalently, after cancellation this corresponds to a vacuum-QM energy plus the MBX many-body/polarization energy, the full permanent QM-MBX electrostatic interaction, and the stationary polarization interaction. The validator checks that MBX’s reported permanent external-field contribution equals $\frac{1}{2}P$ after unit conversion.

4 Sign and unit conventions

Quantity	QUICK convention	MBX handoff convention
Coordinates	bohr internally	Angstrom for MBX ABI
ESP	atomic units, E_h/e	converted by $1/a_0$
EFIELD	atomic units, $E_h/(ea_0)$	converted by $1/a_0^2$
EFG	$G_{ij} = \partial E_i / \partial C_j$	not required for stage-1 energy
Induced dipole	ea_0 in QUICK operator	MBX returns e Angstrom
Energy	hartree in QUICK	MBX reports kcal/mol
Gradient	hartree/bohr, $\partial E / \partial R$	MBX reports kcal/mol/Angstrom

QUICK prints and returns EFG using the field-gradient convention $G_{ij}(\mathbf{C}) = \partial E_i(\mathbf{C}) / \partial C_j$. This is the negative of the electrostatic-potential Hessian convention sometimes also called EFG in quantum-chemistry texts. This convention must be preserved for the gradient/MD stage because derivatives of induced dipoles and fields will use these signs.

The current adapter conversion constants are

$$\phi_{\text{MBX}} = \phi_{\text{QUICK}} / a_0, \quad (18)$$

$$\mathbf{E}_{\text{MBX}} = \mathbf{E}_{\text{QUICK}} / a_0^2, \quad (19)$$

$$\boldsymbol{\mu}_{\text{QUICK}} = \boldsymbol{\mu}_{\text{MBX}} / a_0, \quad (20)$$

where a_0 is expressed in Angstrom. These are the expected conversions for MBX electrostatic units. In the stage-1.5 validation, MBX’s reported half-permanent external-field term agrees with $\frac{1}{2} \sum_s q_s \phi_s^{\text{QM}}$ to better than 2.2×10^{-8} hartree for the small QUICK-MBX water cases.

5 Derivative theory and CPU gradient groundwork

QUICK gradients are derivatives of the total energy with respect to Cartesian coordinates in hartree/bohr. Forces are the negative gradients:

$$\mathbf{F}_I = - \frac{\partial E}{\partial \mathbf{R}_I}. \quad (21)$$

This sign convention is used for both QM nuclei and external classical sites. In the current energy-only interface the converged energy depends on QM nuclear coordinates $\mathbf{R}_A$, the stationary QM density $\mathbf{P}$, MBX real atoms and electrostatic sites $\mathbf{C}_s$, MBX permanent parameters, and MBX induced dipoles. The term-by-term coordinate dependencies are:

- E_{QM} depends on $\mathbf{R}_A$ and on the variationally optimized density $\mathbf{P}$.

- $E_{\text{MBX}}^{\text{internal}}$ depends on MBX real-atom/site coordinates, permanent parameters, polarizabilities, damping, and MBX induced dipoles.
- $E_{\text{QM/MB-MM}}^{\text{perm}} = \sum_s q_s \Phi_s^{\text{QM}}$ depends on QM nuclei, the QM density, and MB permanent-site coordinates.
- The induced-dipole coupling depends on $\mathbf{E}^{\text{QM}}(\mathbf{C}_s)$, MBX induced dipoles, and the MBX polarization model.
- The short-range/dispersion residual depends on the cross QM–MBX coordinates and is currently used as an energy-validation correction, not as a production analytic-gradient term.

5.1 Permanent electrostatic gradients

For a permanent MBX site s , the analytic gradient of the permanent electrostatic energy with respect to the site coordinate is

$$\frac{\partial E_{\text{perm}}}{\partial C_{s,j}} = q_s \frac{\partial \Phi^{\text{QM}}(\mathbf{C}_s)}{\partial C_{s,j}} = -q_s E_j^{\text{QM}}(\mathbf{C}_s). \quad (22)$$

Thus the corresponding force on the MB site is $q_s \mathbf{E}^{\text{QM}}(\mathbf{C}_s)$. This is the first place where QUICK’s EFIELD evaluator enters the QM/MB-MM force path.

For QM nuclei, the same permanent electrostatic coupling is the derivative of the point-charge embedding operator and the QM nuclear–site Coulomb term. QUICK already has this analytic machinery in the external point-charge gradient path. At a converged SCF solution, the ordinary variational cancellation and existing Pulay terms make this the correct analytic permanent-electrostatic slice for QM atom gradients.

5.2 Induced-dipole force terms

The frozen-induced-dipole derivative of the QM field coupling $I = -\sum_s \boldsymbol{\mu}_s \cdot \mathbf{E}^{\text{QM}}(\mathbf{C}_s)$ with respect to an MB site coordinate is

$$\left. \frac{\partial I}{\partial C_{s,j}} \right|_{\boldsymbol{\mu}} = -\sum_i \mu_{s,i} \frac{\partial E_i^{\text{QM}}(\mathbf{C}_s)}{\partial C_{s,j}} = -\sum_i \mu_{s,i} G_{ij}^{\text{QM}}(\mathbf{C}_s). \quad (23)$$

This is the direct reason EFG is needed for analytic gradients and MD. For a fully stationary polarization functional, explicit induced-dipole response terms cancel only if the QUICK operator, MBX energy, and MBX polarization solve are mutually consistent. Production QM/MB-pol gradients still need the MBX derivatives of induced-dipole self/interactions, damping/charge-penetration terms, and real-atom redistribution of site forces.

5.3 MBX force API audit

The local MBX source exports gradient-capable entry points such as `get_energy_g_`, `get_energy_g2_`, and `redistribute_gradients_`. The QUICK adapter now binds the real-atom MBX gradient path and the site-gradient redistribution hook. The MBX gradient unit is kcal/mol/Angstrom, which QUICK converts to hartree/bohr by multiplying by `KCAL_TO_AU` a_0 , with a_0 in Angstrom. The sign is the QUICK gradient sign, $\partial E / \partial R$, as verified against a central finite difference.

This is still not a complete production QM/MB-pol analytic-gradient ABI. The exposed MBX external-call wrapper `set_potential_and_electric_field_on_sites_` sends only ϕ and $\mathbf{E}$ from QUICK to MBX. The underlying MBX system has slots for derivatives of the external field, but the current C/Fortran wrapper does not expose the required field-gradient arrays. Since the external induced-gradient machinery depends on those derivatives, full analytic induced-dipole QM/MB-pol gradients remain blocked until MBX exposes and validates that derivative handoff or provides an equivalent component-gradient API. The accepted local deliverable is therefore a finite-difference total-gradient harness, MBX real-atom gradient parity, and the validated permanent-electrostatic analytic slice.

6 Implementation status

The first staged code patch adds:

- Optional CMake switches `MBX`, `MBX_ROOT`, and `MBX_LIBRARY`. MBX is not vendored into QUICK.
- A thin `quick_mbx_module` adapter around MBX external-call hooks for system initialization, site-coordinate/charge retrieval, external potential/field handoff, MBX energy refresh, and induced dipole retrieval.
- No-I/O QUICK OEPROP evaluators for ESP, EFIELD, and analytic EFG on caller-supplied points.
- A shared ESP OEPROP fix so no-I/O ESP uses the current density matrix, matching the EFIELD/EFG convention and avoiding stale `denseSave` data in library-mode SCF coupling.
- A CPU AO point-charge operator for MBX permanent site charges and for reuse by the finite-difference dipole operator.
- A CPU AO point-dipole operator, `add_point_dipole_field_operator`, currently implemented as a centered finite difference of the existing point-charge OEI path.
- Serial SCF coupling hooks and MPI root-rank MBX handoff with induced-dipole broadcast. The MPI operator follows QUICK’s shell-pair distribution and final Fock reduction style.
- A post-SCF MBX energy refresh and QUICK total-energy hook.
- A corrected energy decomposition that prints the raw MBX energy, the accepted additive MBX correction, permanent total/classical/ electronic terms, induced total/classical/electronic terms, and the MBX half-permanent residual check.
- Library-mode API entry points `setQuickMBXSystem` and `setQuickMBXWaterSystem`, plus `clearQuickMBXSystem`; `deleteQuickJob` also finalizes any active MBX system.
- MBX force-adapter entry points:

```
quick_mbx_get_site_data
quick_mbx_get_induced_dipoles_au
quick_mbx_set_real_coordinates
quick_mbx_get_real_atom_gradient
quick_mbx_redistribute_site_gradient
```

These are guarded by the optional MBX build path and do not change non-MBX builds.

- A public reference API, `getQuickMBXEnergyGradientsFD`, for tiny water-environment finite-difference total QUICK–MBX gradients. This API is intended for validation and CPU smoke dynamics, not production analytic MD.
- Optional water and methane dimer validation drivers, `test-api-mbx-dimer-scan` and `test-api-mbx-ch4-dimer-scan`, for laptop-scale publication checks against MBX standalone calculations.
- A CPU derivative validation driver, `test-api-qmmm-perm-grad`, which compares QUICK’s analytic permanent point-charge gradients against central finite differences and checks analytic EFG against finite-difference EFIELD. The driver is MPI-aware, so `test-api-qmmm-perm-grad.MPI` performs a genuine two-rank QUICK API calculation.
- Stage-4 CPU gradient smoke drivers:

```
test-api-mbx-gradient-parity
test-api-mbx-total-fd-gradient
test-api-mbx-fd-md-smoke
test-api-mbx-gradient-decomp
```

- An experimental `MBX_QMMM` keyword guarded to energy-only jobs.

Current limitations:

- MBX system initialization exists in the adapter and a convenience API is available for MB-pol water. General production input-file syntax for MBX systems is still not complete.
- MBX’s Fortran external-call ABI does not expose a site-count query, so the caller must provide the expected number of electrostatic sites when initializing the adapter. For MB-pol water this is four sites per monomer.
- The stage-1.5 energy bookkeeping is accepted for the current energy-only target and validated by the MBX half-permanent residual check. Broader production validation should still add more geometries before treating the interface as chemically production-ready.
- The AO electric-field operator should be replaced by direct analytic recurrence contraction after the interface is validated.
- Full induced-dipole and MB-pol analytic gradients are not yet implemented. The current production analytic derivative claim is limited to the permanent electrostatic point-charge slice and OEPROP derivative ingredients. The full total QUICK–MBX gradient is available only as a finite-difference reference path for tiny CPU tests.

7 Validation

The stage-1 validation driver is

```
python3 validation/quick_mbx/validate_quick_mbx_stage1.py
```

It checks serial saved references for ESP/EFIELD/EFG density-surface cases, serial-vs-MPI agreement for the same properties, and analytic EFG against the finite-difference EFG reference. In the current local run, all QUICK checks passed and the analytic-vs-numerical EFG maximum absolute difference was 1.0×10^{-9} . The current stage-1.5 local validation results are summarized below.

Check	Result	Tolerance
ESP/EFIELD/EFG saved refs	max diff 0	2.0×10^{-7}
ESP/EFIELD/EFG serial vs MPI	max diff 0	2.0×10^{-7}
Analytic EFG vs numerical EFG	1.0×10^{-9}	2.0×10^{-7}
MBX 001_mbp01 parity	2.0×10^{-9} kcal/mol	1.0×10^{-8}
QUICK-MBX 1-water smoke	serial/MPI diff $1.85 \times 10^{-13} E_h$	1.0×10^{-8}
QUICK-MBX 2-water smoke	serial/MPI diff $9.95 \times 10^{-14} E_h$	1.0×10^{-8}
MBX half-permanent residual	$2.20 \times 10^{-8} E_h$	1.0×10^{-6}

The same driver contains optional MBX-dependent checks. With `--mbx-home` and, when the example tree is separate from the install, `--mbx-source`, it runs MBX’s 001_mbp01 example through the installed `single_point` executable and compares the output to MBX’s expected reference. The Python example remains a fallback path.

7.1 CPU derivative validation

The stage-3 derivative probe uses a water monomer as the QM subsystem and three fixed MB-pol-like permanent sites for the second monomer: two positive hydrogen sites and one negative massless site on the water bisector. This deliberately avoids MBX force APIs so that the available analytic slice can be tested in isolation. The calculation uses HF/STO-3G, a central finite-difference displacement of 10^{-3} bohr, and the public QUICK API routine `getQuickEnergyGradients`. The same driver calls `getQuickOEPROP` to compare analytic EFG with central finite differences of EFIELD.

Check	Serial	Two-rank MPI
QM atom max	3.9855×10^{-7}	3.9847×10^{-7}
QM atom RMS	1.8322×10^{-7}	1.8320×10^{-7}
MM site max	1.6504×10^{-9}	1.6717×10^{-9}
MM site RMS	8.8055×10^{-10}	8.9742×10^{-10}
EFG vs FD EFIELD max	1.6494×10^{-7}	1.6494×10^{-7}
EFG vs FD EFIELD RMS	4.8206×10^{-8}	4.8206×10^{-8}

All gradient rows are permanent-electrostatic errors in hartree/bohr, and all EFG rows are analytic-vs-finite-difference errors in atomic units. During development this probe exposed an important validation detail: OEPROP includes QUICK external point charges in the returned field and field gradient. Therefore, placing OEPROP probes exactly on the validation point charges creates singular self-field terms. The final probe uses nearby non-coincident points for the EFG derivative check. This does not change the production QUICK-MBX handoff, where MBX sites are not registered as QUICK external point charges when ESP/EFIELD are evaluated for MBX.

7.2 Stage-4 QUICK-MBX finite-difference gradient and MD smoke

The stage-4 CPU gradient target adds a deliberately conservative reference path before claiming analytic QM/MB-pol forces. The adapter first checks MBX’s own real-atom analytic gradients through `get_energy_g_`. Then `getQuickMBXEnergyGradientsFD` computes central finite-difference

total QUICK-MBX gradients by repeatedly calling the accepted energy-only self-consistent interface. This route uses no `GRADIENT` keyword with `MBX_QMMM`; it is an API reference path for tiny systems.

The local smoke system is one QUICK water monomer interacting with one MBX MB-pol water monomer at the fixed dimer geometry used throughout these notes. The QUICK method is HF/STO-3G with tightened integral cutoffs. This is not a chemical benchmark; it is a sign, unit, bookkeeping, and repeated API-call stability test for later PBE0/PBE and analytic-force work.

Check	Result	Notes
MBX real-atom gradient parity max	3.6013×10^{-7}	hartree/bohr
MBX real-atom gradient parity RMS	1.6530×10^{-7}	hartree/bohr
Total FD gradient step sensitivity	5.1074×10^{-5}	max difference, 10^{-3} vs 2×10^{-3} bohr
Total FD net gradient norm	2.2335×10^{-5}	hartree/bohr
Total FD max gradient component	5.5062×10^{-2}	hartree/bohr
Two-step FD-MD smoke drift	6.9718×10^{-8}	hartree over two 0.10 fs steps

The MD smoke executable uses velocity-Verlet with forces $-\partial E/\partial R$ from the finite-difference total gradient. All coordinates passed through the public API are Angstrom; all finite differences and gradients are reported in bohr-based atomic units. The small energy drift is a useful local sanity check that the repeated QUICK-MBX calls, force sign, mass conversion, and energy expression are self-consistent for a tiny system. It should not be interpreted as a production MD performance or timestep recommendation.

7.3 Stage-4.5 gradient decomposition diagnostic

The next development diagnostic asks how much of the MBX-side total finite-difference gradient can already be recovered from analytic pieces. The probe computes, on the same one-water/one-water smoke geometry:

- the total QUICK-MBX gradient by central finite differences of the accepted self-consistent energy expression;
- the MBX real-atom internal gradient from `get_energy_g_` without an external QUICK field;
- the QUICK permanent electrostatic site-gradient term $-q_s \mathbf{E}^{\text{QM}}(\mathbf{C}_s)$, redistributed from MBX sites to real atoms;
- the frozen-induced site-gradient term $-\sum_i \mu_{s,i} G_{ij}^{\text{QM}}(\mathbf{C}_s)$, also redistributed to real atoms.

This is a diagnostic decomposition, not a production gradient. It uses the available QUICK EFIELD/EFG convention and MBX redistribution hook while leaving the unavailable induced-dipole response and external-field bookkeeping derivatives in the residual.

MBX-side model	Max residual	RMS residual
MBX internal only	3.2401×10^{-3}	1.4613×10^{-3}
Internal + permanent site gradient	7.0280×10^{-4}	3.9001×10^{-4}
Internal + permanent + frozen-induced	8.3468×10^{-4}	4.3899×10^{-4}

All residuals are relative to the MBX-atom components of the total finite-difference gradient and are reported in hartree/bohr. The maximum MBX-atom finite-difference component for this

geometry is 1.0558×10^{-2} hartree/bohr. The permanent site-gradient term therefore removes most of the missing MBX-side force. The frozen-induced term is small here and does not close the residual by itself, which is consistent with the MBX audit: production induced gradients require the external field-gradient/response path inside MBX, not only a frozen $\mu \cdot G$ contraction in QUICK.

7.4 Water dimer PES

The Lambros-style dimer-scan validation script evaluates the water-dimer residual in Eq. 11 on the same 20 geometries used for the QUICK and QUICK-MBX calculations. The MBX baseline is generated by copying the MB-pol `mbx.json` file and setting `ignore_2b_poly = [[h2o,h2o]]` and `ignore_dispersion = [[h2o,h2o]]`. Thus the baseline retains the MB-pol permanent electrostatics and polarization terms while disabling the cross h2o/h2o two-body short-range polynomial and dispersion terms.

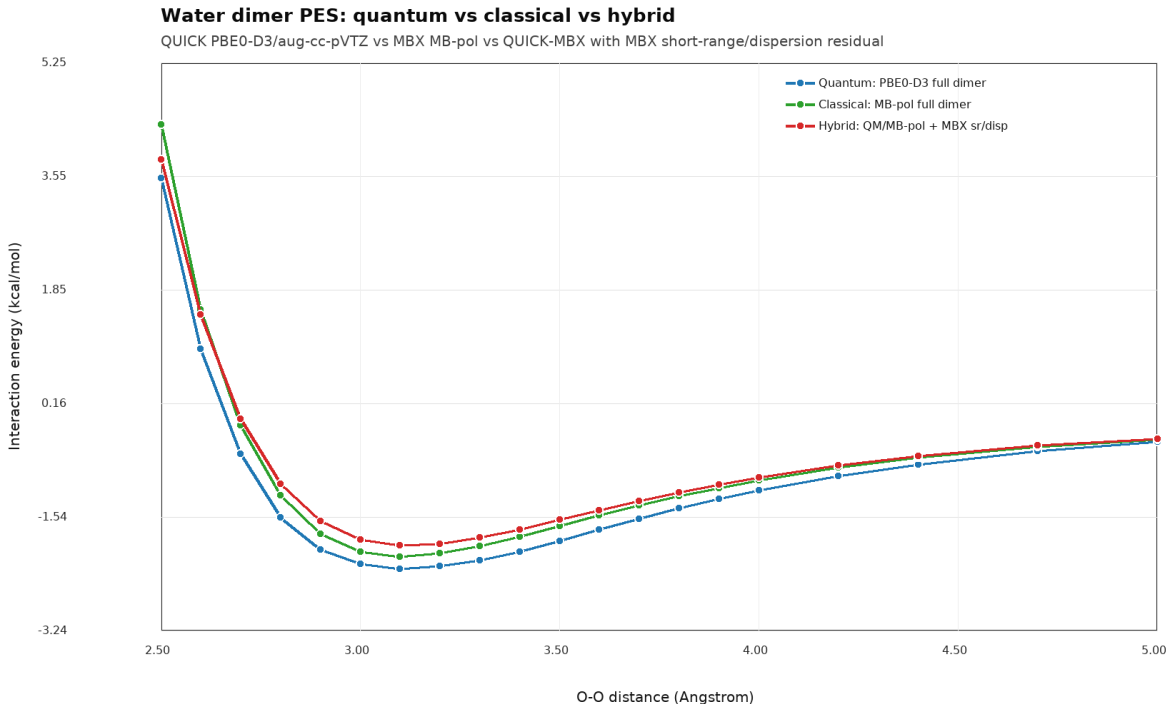


Figure 1: Water-dimer interaction-energy scan on identical XYZ points. The hybrid curve is the complete current QM/MB-pol validation energy: QUICK PBE0-D3 for monomer *A*, MBX MB-pol electrostatics and polarization for monomer *B*, plus the MBX h2o/h2o short-range and dispersion residual.

Representative points from the 20-point PBE0-D3/aug-cc-pVTZ scan are:

R_{OO} (Å)	PBE0-D3	MB-pol	raw hybrid	$\Delta E_{sr/disp}$	corrected hybrid
2.50	3.525636	4.318132	-11.442793	15.243055	3.800262
3.10	-2.309084	-2.128275	-3.089691	1.123305	-1.966385
5.00	-0.418826	-0.385009	-0.352831	-0.017670	-0.370501

All energies in this table are interaction energies in kcal/mol. The raw column is retained only as a diagnostic decomposition, not as a complete QM/MM method. The corrected hybrid now has

the expected qualitative behavior: a repulsive short-range wall, an attractive near-minimum point, and a long-range tail that decays with the full QM and full MB-pol curves. The corrected hybrid does not yet prove quantitative production accuracy, but it removes the unphysical flipped curve observed with electrostatic/polarization coupling alone.

7.5 MB-pol-optimized MB-PBE water-dimer scan

MBX also provides an `mbpbe` water model, effectively an MB-pol-like potential trained against a PBE-D3 reference. This makes it a useful classical partner for a PBE-D3/MB-PBE QUICK-MBX validation. The local MBX installation contains an `optimize` executable, but that executable is unconstrained and would relax all scan points toward ordinary dimer minima. For a potential-energy scan, the O–O separation must remain fixed.

A lightweight constrained MBX-gradient optimizer was therefore added:

- `optimize_water_dimer_mbx_scan.py` calls MBX analytic gradients through the local MBX Python binding, projects the gradient onto the tangent space of the fixed O(1)–O(4) distance, and relaxes the remaining Cartesian degrees of freedom with a small projected L-BFGS optimizer.
- `run_mbpbe_water_scan.py` consumes that XYZ stack and evaluates five fixed-geometry interaction curves: full QUICK PBE-D3/aug-cc-pVTZ, full MB-pol, full MB-PBE, corrected PBE-D3/MB-pol, and corrected PBE-D3/MB-PBE.

The MB-PBE hybrid required one adapter-level ABI fix. MBX’s legacy `initialize_system` entry point uses fixed five-character C buffers for monomer names, which is enough for `h2o` but not for `mbpbe` plus a null terminator. QUICK now calls MBX’s existing `initialize_system.py` entry point, which accepts `char**` name arrays and therefore passes `mbpbe` without modifying MBX.

The dimer residual convention is the same as Eq. 11. For MB-pol the residual baseline sets `ignore.2b.poly = [[h2o,h2o]]` and `ignore.dispersion = [[h2o,h2o]]`. For MB-PBE the corresponding baseline sets `ignore.2b.poly = [[mbpbe,mbpbe]]` and `ignore.dispersion = [[mbpbe,mbpbe]]`. The corrected hybrid curves therefore include the MBX short-range and dispersion cross residual rather than showing only the raw electrostatic/polarization coupling.

R_{OO}	PBE-D3	MB-pol	MB-PBE	PBE-D3/MB-pol	PBE-D3/MB-PBE
2.50	-2.650359	-3.132446	-2.657764	-2.349579	-2.536994
2.90	-5.477824	-4.988697	-5.392890	-4.739664	-5.940941
5.00	-0.903218	-0.869014	-0.863688	-0.847918	-0.904959

All energies in the table are interaction energies in kcal/mol. The 20 MB-pol constrained geometries were generated in about 13 seconds and all points satisfied the projected-force acceptance criterion (`max force` < $5\text{e-}4$ kcal mol^{−1} Å^{−1} or `RMS force` < $2.5\text{e-}4$ kcal mol^{−1} Å^{−1}). On these relaxed geometries, MB-PBE tracks the full PBE-D3 curve more closely than MB-pol. The target PBE-D3/MB-PBE hybrid is slightly overbound near the minimum but has a smooth well and a long-range tail that nearly coincides with PBE-D3. The PBE-D3/MB-pol curve is retained as a reference hybrid with the original MB-pol water model.

The production commands were:

```
export QUICK_BUILD=/private/tmp/quick-build-mbx-focktest
export CODEX_PY=/Users/etiennepalos/.cache/codex-runtimes/\
codex-primary-runtime/dependencies/python/bin/python3
```

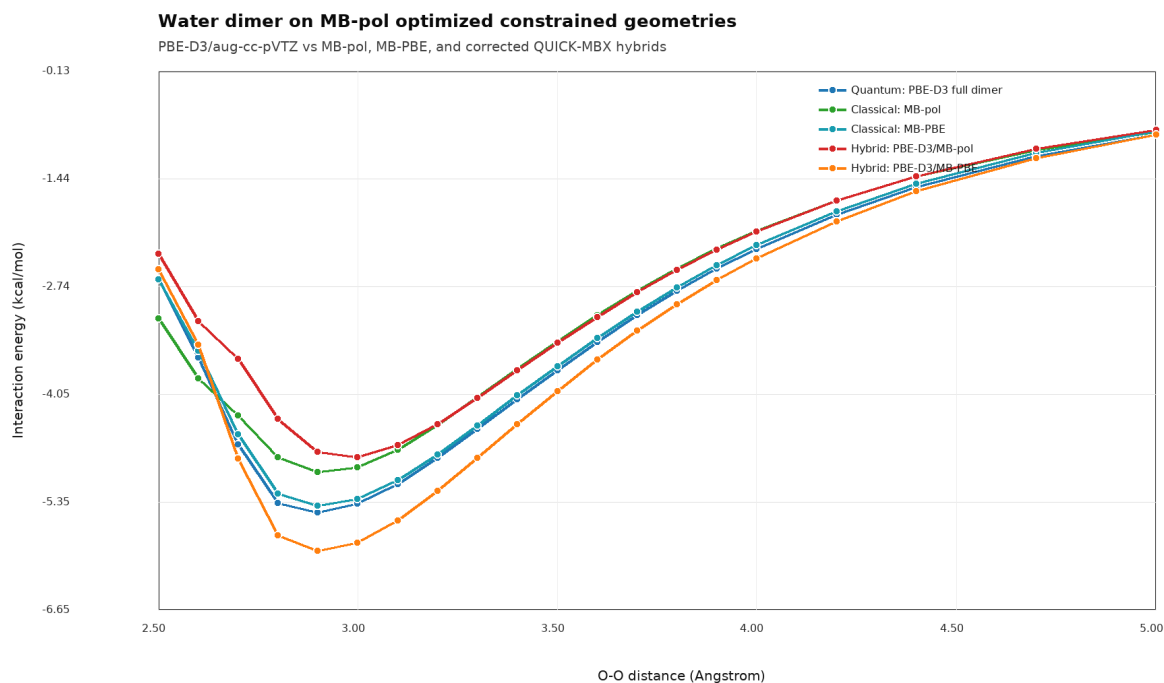


Figure 2: Twenty-point water-dimer interaction-energy scan on MB-pol optimized constrained geometries. Each point fixes R_{OO} and relaxes the remaining coordinates with MB-pol before evaluating QUICK PBE-D3/aug-cc-pVTZ, full MB-pol, full MB-PBE, corrected PBE-D3/MB-pol, and corrected PBE-D3/MB-PBE energies.

```

cmake --build $QUICK_BUILD \
  --target quick test-api-mbx-water-xyz-scan --parallel 8

python3 validation/quick_mbx/optimize_water_dimer_mbx_scan.py \
  --max-steps 600 --force-tol 5.0e-4

$CODEX_PY validation/quick_mbx/run_mbpbe_water_scan.py \
  --scan-exe $QUICK_BUILD/src/test-api-mbx-water-xyz-scan \
  --xyz validation/quick_mbx/water_dimer_mbpol_optimized_20pt.xyz \
  --csv validation/quick_mbx/water_dimer_mbpolopt_pbe_d3_mbpbe_20pt.csv \
  --output validation/quick_mbx/water_dimer_mbpolopt_pbe_d3_mbpbe_20pt.png

```

7.6 Troubleshooting the MB-PBE hybrid overbinding

The first MB-PBE hybrid scan is scientifically useful because full PBE-D3 and full MB-PBE agree closely, while the corrected PBE-D3/MB-PBE hybrid is still modestly overbound near the minimum. This rules out a simple “MB-PBE is the wrong classical reference” explanation. The current corrected hybrid interaction curve is

$$E_{\text{hyb}}^{\text{current}} = E_{\text{raw}}^{\text{QM/MBPBE}} + \Delta E_{\text{sr/disp}}^{\text{MBPBE}}, \quad \Delta E_{\text{sr/disp}}^{\text{MBPBE}} = E_{\text{full}}^{\text{MBPBE}} - E_{\text{base}}^{\text{MBPBE}}. \quad (24)$$

Here $E_{\text{raw}}^{\text{QM/MBPBE}}$ is the self-consistent QUICK/MBX electrostatic-polarization baseline, while $E_{\text{base}}^{\text{MBPBE}}$ is the MB-PBE electrostatic-polarization baseline obtained by disabling the water–water 2B polynomial and dispersion. The MB-PBE short-range and dispersion residual was trained to correct $E_{\text{base}}^{\text{MBPBE}}$, not the modified baseline $E_{\text{raw}}^{\text{QM/MBPBE}}$. The diagnostic baseline-mismatch term is therefore

$$\delta_{\text{base}} = E_{\text{raw}}^{\text{QM/MBPBE}} - E_{\text{base}}^{\text{MBPBE}}. \quad (25)$$

Subtracting this term gives a baseline-aligned diagnostic curve,

$$E_{\text{hyb}}^{\text{aligned}} = E_{\text{hyb}}^{\text{current}} - \delta_{\text{base}}. \quad (26)$$

For this two-water validation case the aligned diagnostic is numerically equivalent to replacing the cross electrostatic-polarization baseline by the MB-PBE baseline before adding the MB-PBE residual. It is therefore not yet a production QM/MM energy model; it is a bookkeeping diagnostic that isolates why the current hybrid overbinds.

Three-point Hamiltonian-scale diagnostics support the same conclusion. Removing the induced-dipole Fock operator changed the 2.90 Å error only from −0.46 to −0.42 kcal/mol, so induced coupling is not the dominant source. Removing the permanent-site Fock operator moved the 2.90 Å error to +0.28 kcal/mol, and a half-scale permanent operator gave −0.27 kcal/mol while worsening the short-range point. Thus a constant scale is not an acceptable production fix. The production path should instead implement a physically motivated permanent-electrostatics damping or charge-penetration correction, or obtain an MBX component API for a QM/MM-specific short-range residual that is trained or evaluated against the actual QM/MB electrostatic-polarization baseline.

7.7 Small pair scans beyond water

The same residual workflow was extended to three additional small-pair checks. Methane–methane uses the MBX 040_ch4-ch4_mb-nrg_2bnb example with five MBX sites per methane. The residual baseline disables `ignore_2b_poly = [[ch4,ch4]]` and `ignore_dispersion = [[ch4,ch4]]`.

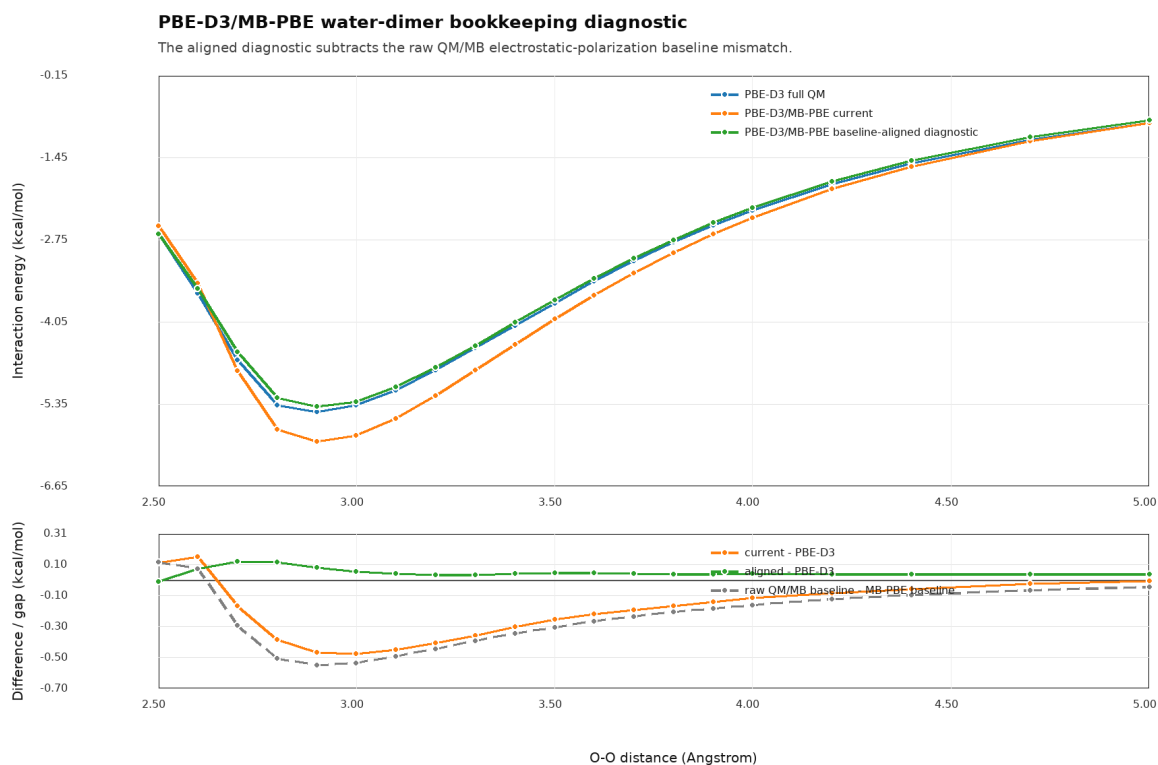


Figure 3: Bookkeeping diagnostic for the MB-PBE water-dimer scan. The current PBE-D3/MB-PBE hybrid overbinding tracks the raw QM/MB electrostatic-polarization baseline mismatch relative to the MB-PBE baseline. Removing that mismatch reduces the 20-point RMS error against PBE-D3 from 0.269 to 0.060 kcal/mol.

The H₂O-CH₄ scan uses the MBX 042_ch4-h2o_mb-nrg_2bnb example with `ignore_2b_poly = [[ch4,h2o]]` and `ignore_dispersion = [[ch4,h2o]]`. The Cl-/H₂O scan uses the MBX 022_h2o-cl_mb-nrg_2bnb example with `ignore_2b_poly = [[cl-,h2o]]` and `ignore_dispersion = [[cl-,h2o]]`.

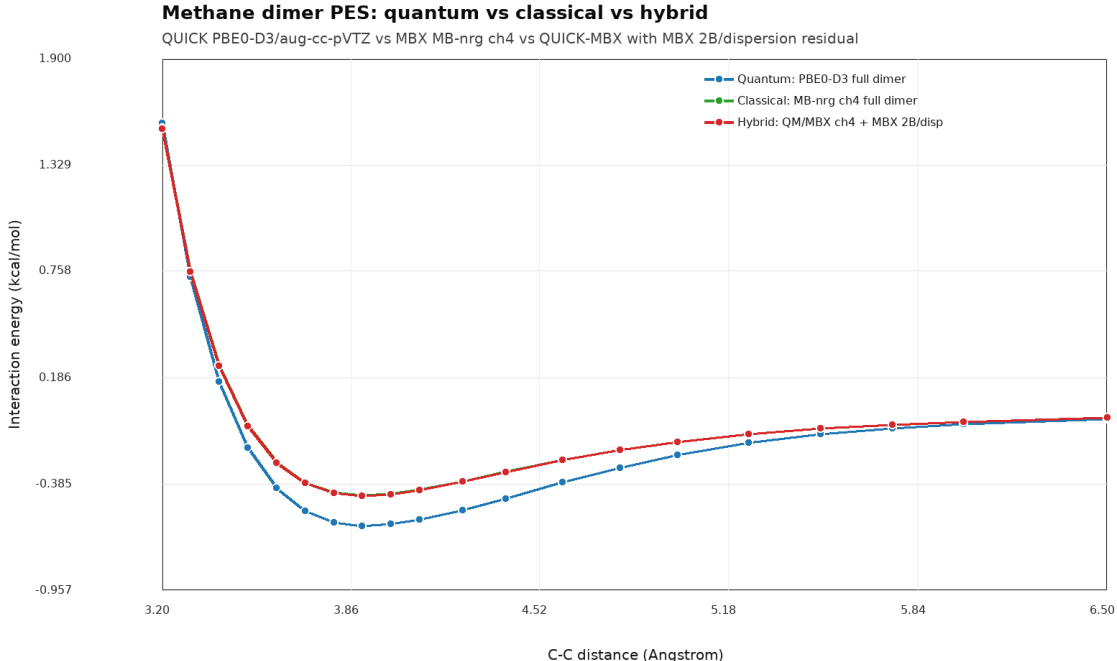


Figure 4: Twenty-point methane-dimer scan. The corrected hybrid curve is QUICK PBE0-D3/aug-cc-pVTZ on monomer A, MBX MB-nrg methane on monomer B, plus the isolated ch₄/ch₄ two-body and dispersion residual.

System	QM fragment	Points	Residual	$R_{\min}$	PBE0-D3	MBX	Hybrid
CH ₄ -CH ₄	CH ₄	20	2B/disp	3.90	-0.610	-0.447	-0.448
H ₂ O-CH ₄	H ₂ O	10	2B/disp	3.60	-0.763	-0.560	-0.559
H ₂ O-CH ₄	CH ₄	10	2B/disp	3.60	-0.763	-0.560	-0.608
Cl-/H ₂ O	Cl-	10	2B/disp	3.20	-14.588	-14.095	-12.183
O ₂ /H ₂ O	triplet O ₂	10	none	6.00	-0.033	N/A	6.769

Table 1: Representative minima from the additional scans. Distances are in Angstrom and energies are interaction energies in kcal/mol. The “Hybrid” column is the corrected hybrid where an MBX cross residual can be isolated; for O₂/H₂O it is the electrostatic/polarization-only diagnostic.

The CH₄-CH₄ and H₂O-CH₄ scans give the desired qualitative publication behavior: a short-range wall, an attractive minimum, and a tail that decays toward zero. The corrected hybrid follows the corresponding MBX curve closely for methane and closely tracks the mixed H₂O-CH₄ MBX reference after the residual is added. The Cl-/H₂O scan also converges and has the expected well and tail, but the first short-range point is too repulsive relative to both PBE0-D3 and the MBX ion-water model. This should be revisited before using ionic systems as an acceptance benchmark. The triplet O₂/H₂O case verifies the unrestricted open-shell path and MBX water coupling, but

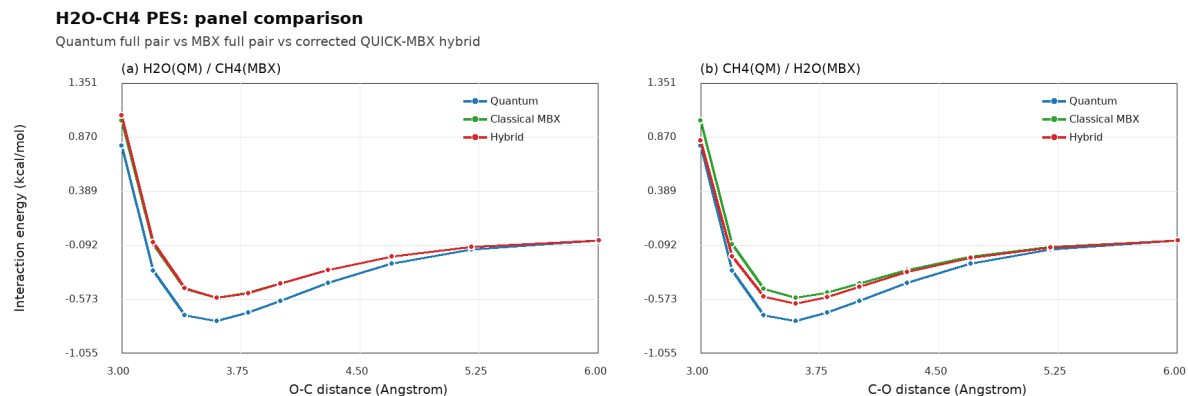


Figure 5: Ten-point H2O-CH4 role-swap scan. Panel (a) treats H2O as QM and CH4 as MBX; panel (b) treats CH4 as QM and H2O as MBX. Both panels use the same MBX ch4/h2o two-body and dispersion residual.

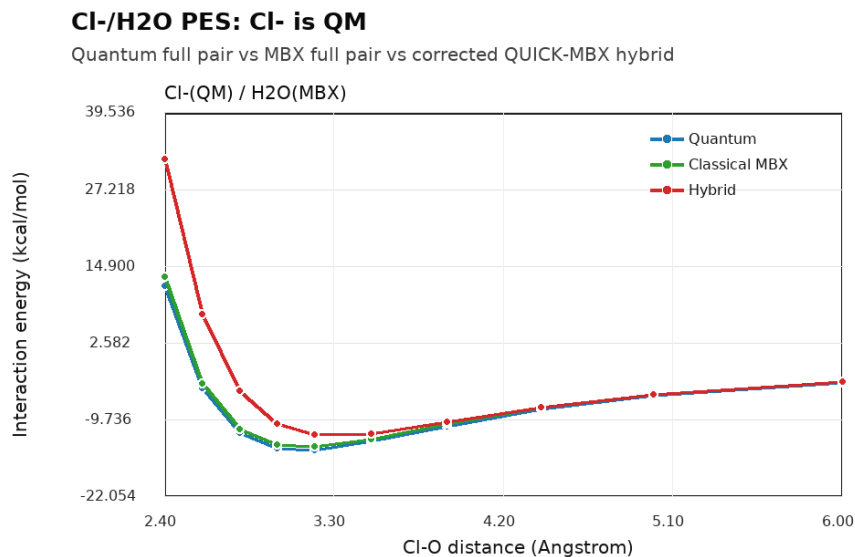


Figure 6: Ten-point Cl-/H2O scan with Cl- as the QM subsystem. The corrected hybrid uses the MBX cl-/h2o short-range and dispersion residual. The short-range wall is over-repulsive at the first point, so this ionic case is a useful stress test rather than a quantitative acceptance result.

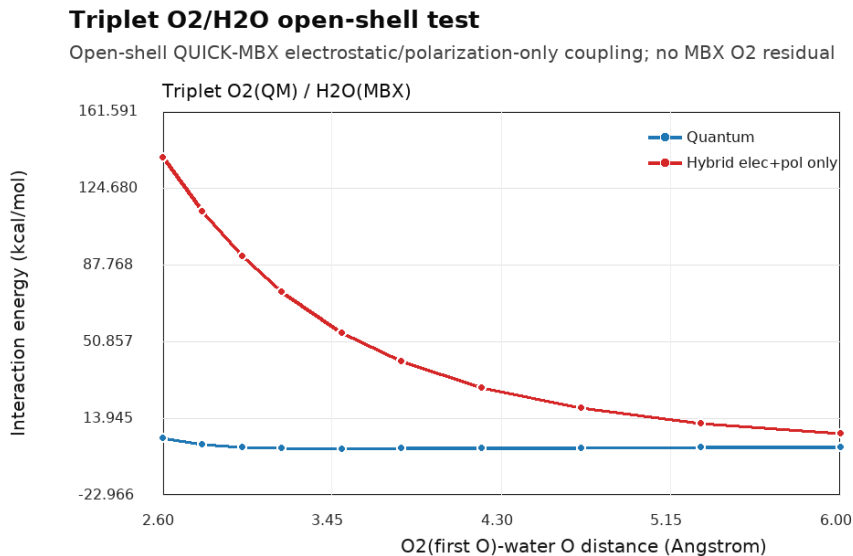


Figure 7: Ten-point triplet O2/H2O open-shell smoke test. QUICK runs the O2-containing calculations as unrestricted triplets. No MBX O2 model or O2/H2O residual is available in the local MBX tree, so the red curve is electrostatic/polarization coupling only and is not a complete QM/MM method.

because no O2 cross residual is available it is not a complete chemical validation.

7.8 Trimer roadmap

A trimer scan was not attempted in this resource window. For a future water-trimer validation with monomer A in QUICK and monomers B, C in MBX, the cross residual should be isolated explicitly rather than silently mixing MB-only and QM/MB terms. A practical MBX-derived residual is

$$\Delta E_{\text{cross}}^{ABC} = [E_{\text{full}}^{ABC} - E_{\text{elec+pol}}^{ABC}] - [E_{\text{full}}^{BC} - E_{\text{elec+pol}}^{BC}]. \quad (27)$$

The first bracket contains all non-electrostatic residual terms in the trimer, while the second removes the purely MBX B - C residual already belonging to the MBX internal environment. A 2B-only correction should disable only h2o/h2o two-body polynomial and dispersion terms. Any 3B short-range correction must be isolated and reported separately by also controlling the h2o/h2o/h2o three-body polynomial term.

When QUICK is configured with `-DMBX=TRUE`, CMake also builds optional `test-api-mbx` and `test-api-mbx.MPI` smoke executables. These initialize one and two MB-pol waters through `setQuickMBXWaterSystem`, run fixed-geometry energy-only QUICK SCF calculations with `MBX_QMMM`, check SCF convergence, and print parseable total energies. The validation driver runs them automatically when the executables are present or when paths are supplied with `--quick-mbx-smoke` and `--quick-mbx-smoke-mpi`. The next validation cases should be:

- QUICK-generated ESP/EFIELD at MBX sites compared to a controlled point-charge reference.

- More fixed-geometry QM/MB-pol water monomer/dimer geometries, kept small enough for routine laptop validation.

8 Roadmap toward gradients and MD

EFG is not required for the first energy-only QM/MB-pol target, but it is needed for analytic gradients and MD. Stage 4 provides a CPU finite-difference total-gradient reference and a tiny MD smoke test. Stage 4.5 adds a decomposition diagnostic showing that MBX internal forces plus the QUICK permanent site-gradient term recover most of the MBX-side total finite-difference gradient for the tiny water test. Production analytic QM/MB-pol gradients still require:

- Derivatives of QUICK ESP and EFIELD at MBX sites with respect to QM nuclear coordinates, including direct analytic AO field-operator derivatives rather than finite differences of point-charge operators.
- A MBX external-derivative ABI that accepts the QUICK field-gradient tensor and any induced-dipole response data needed by MBX’s external induced-gradient machinery.
- Component-consistent gradients for the same energy bookkeeping used in Eq. 16 and for any MBX short-range/ dispersion residual added to a corrected hybrid validation energy.
- Consistent redistribution of MBX site gradients to real atoms, validated for virtual sites and lone-pair sites.
- Explicit tests that the QUICK field-gradient convention $G_{ij} = \partial E_i / \partial C_j$ is used throughout.

The next QUICK-side development step is to promote the permanent $-q_s \mathbf{E}^{\text{QM}}(\mathbf{C}_s)$ MBX-site gradient and MBX redistribution path from diagnostic code into an analytic gradient component, with finite-difference regression tests on fixed water geometries. The highest-value QUICK-side refactor after that is a direct analytic AO field-operator derivative contraction that shares recurrence tables with EFIELD/EFG. The highest-value MBX-side interface improvement is an external-call wrapper that exposes and validates external field-gradient handoff without requiring QUICK to vendor or patch MBX locally.

References

- Lambros, E.; Lipparini, F.; Cisneros, G. A.; Paesani, F. A many-body, fully polarizable approach to QM/MM simulations. *J. Chem. Theory Comput.* 2020, 16, 7462–7472. DOI: 10.1021/acs.jctc.0c00932.
- The OpenMMPol Library for Polarizable QM/MM Calculations of Properties and Dynamics. 2024. <https://arxiv.org/abs/2401.14691>.
- MBX source and documentation. <https://github.com/paesani-lab/MBX>.